Homework 4: Machine Translation

CS 1470/2470

Due November 5, 2019 at 11:59pm

1 CS2470-only Questions

1. What requires more parameters: single or multi-headed attention? Explain. Does this mean one necessarily trains faster than another? (3-5 sentences)

   Answer: It is really hard to say which model has more variables since there is no information on the structure and size of the model. Given all the model other than the attention head, if set the size of each weight head as embedding size/3 * embedding size and there are 3 heads, I would say the multi-headed model has more parameters to train, and to train such a network would cost more time. The multi-headed model needs to train a weight matrix after concatenating all 'z' vector, however there is no such a matrix in single head model. And this is the reason why multi-headed model is typically slower.

2. For the last homework, we asked you to consider convolutional architectures for language modeling, and to weigh their trade-offs against RNN-based architectures.

   Transformers can also be used for language modeling. In fact, they are current state of the art.
   (see https://openai.com/blog/better-language-models/).

   How are transformer-based language models similar to convolution? What makes them more suited for language modeling? (5-10 sentences)

   Answer: In terms of model structure, the Transformer model has both decoder and encoder part, and CNN model could also has the same part. For each block, the self-attention part works very similar as the filter part: extract each word information in a sentence and adjust weight matrices accordingly. I would say the easiness to train really makes them suitable for language model since the model training data could as large as hundreds of GB, which makes most of recurrent model extremely hard and slow to train. But this kind of simple neural network is almost trivial to parallelize.